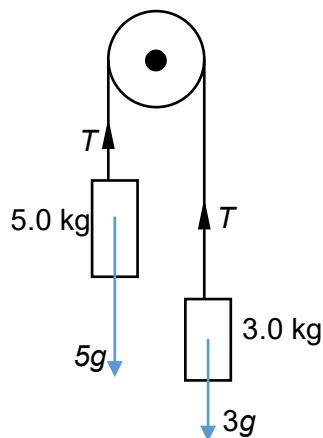


forces acting on the system when the two objects collide.

5 A

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By considering the free body diagrams of the 5.0 kg and 3.0 kg masses:

$$5g - T = 5a$$

$$T - 3g = 3a$$

$$\therefore 2g = 8a$$

$$a = \frac{1}{4}g$$

$$= 2.5 \text{ m s}^{-1} \text{ (to 2 s.f.)}$$